Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Deok-Ho Kim	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University School of Medicine			
Address	Rm. 724, Ross Research Building 720 Rutland Avenue Baltimore, MD 21205			

2. Technology Overview

Title of Proposed Technology	Advanced Flow Electroporation–Based Manufacturing Platform for High-Quality Adeno-Associated Virus Production			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Gene Pharmaceuticals	Bio-processes/ Bio-equipment	Bio-process Technologies
Concept of Proposed Technology	Adeno-associated virus (AAV) vectors are a core enabling technology for gene therapies targeting rare and intractable diseases. Despite rapid clinical expansion, large-scale AAV manufacturing remains limited by **low process efficiency, high plasmid consumption, and poor vector quality**, particularly the high proportion of empty capsids generated during conventional production. Current chemical transfection methods suffer from heterogeneous intracellular plasmid delivery and poor control of multi-plasmid stoichiometry. While electroporation enables efficient cytoplasmic plasmid delivery, its use in industrial AAV manufacturing has been constrained by **electroporation-induced cellular damage**, including thermal stress, membrane disruption, mitochondrial dysfunction, and loss of post-transfection cell viability and proliferation. The proposed technology introduces a **next-generation flow electroporation platform** designed to overcome these limitations by integrating: • Thermal-stress–mitigated flow electroporation architectures • Cytoprotective process conditions that preserve post-transfection cell fitness • Uniform intracellular delivery of multiple plasmid DNAs (pDNAs) required for AAV			

	production By preserving cell proliferation and enabling balanced expression of vector components, the platform aims to **reduce plasmid input requirements and significantly decrease AAV empty capsid formation**, enabling a scalable, biologically robust, and commercially viable AAV manufacturing process.
Rationale for Joint R&D Project	Development of an advanced flow electroporation–based AAV manufacturing process requires **simultaneous expertise in equipment engineering, cell process optimization, and viral vector biology**, which are rarely consolidated within a single organization. In particular, rigorous validation of post-electroporation cell fitness, nuclear plasmid entry dynamics, and vector assembly quality requires **independent biological evaluation and advanced AAV production infrastructure**, typically available only at a limited number of overseas research institutions. International joint R&D is therefore essential to: • Reduce early-stage technical and biological risk • Enable objective benchmarking against conventional transfection methods • Accelerate maturation from laboratory feasibility to manufacturing-relevant validation Within this collaboration, the research institute provides independent biological validation and benchmarking, while the Korean SME partner leads core process development, system integration, scale-up, and commercialization.
Key Words	1. Flow Electroporation 2. AAV Manufacturing 3. Empty Capsid Reduction 4. Multi-pDNA Delivery 5. Cell Fitness Preservation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	**Overall Objective** To develop and validate a **next-generation flow electroporation–based AAV manufacturing process** that preserves post-transfection cell fitness, enables uniform multi-plasmid delivery, improves plasmid utilization efficiency, and reduces empty capsid formation, advancing the technology from **TRL 4 to TRL 6**. **Specific Objectives** • Minimize electroporation-induced thermal and electrical cellular damage • Preserve post-electroporation cell viability and proliferation within defined limits • Achieve uniform intracellular and nuclear delivery of multiple plasmids • Maintain AAV production yield using reduced plasmid input • Substantially reduce AAV empty capsid formation			

	• Validate reproducibility, robustness, and scalability in a relevant AAV production environment
R&D Target Specifications	Key quantitative targets include: • **Post-Electroporation Cell Viability**: ≤ 10% reduction compared to untreated control cells • **Post-Electroporation Cell Proliferation**: ≤ 20% reduction in proliferation rate • **Multi-pDNA Expression Uniformity**: Highly homogeneous cytosolic and nuclear expression with reduced variability relative to chemical transfection • **Plasmid Utilization Efficiency**: Equivalent or lower amount of plasmid input required compared to PEI-based transfection while maintaining AAV yield • **AAV Production Yield**: Equivalent or higher vector genome (vg) yield relative to PEI-based methods • **AAV Empty Capsid Ratio**: ≤ 30% in final AAV product • **Technology Readiness**: Advancement from TRL 4 to TRL 6
R&D Plan	The R&D plan follows a **phased optimization and validation strategy** designed to isolate electroporation-specific effects and progressively demonstrate manufacturing relevance. Phase 1: Electroporation Optimization Using Reporters • Optimize flow electroporation parameters while excluding plasmid-induced toxicity • Evaluate cell viability, proliferation, and cytosolic delivery uniformity using multi-color DNA(or mRNA) reporters • Identify conditions achieving preserved cell fitness and uniform delivery Phase 2: Nuclear Delivery Optimization Using Plasmid DNA • Replace reporters with fluorescent plasmid DNAs • Optimize nuclear entry and expression balance of multiple plasmids • Correlate cell cycle progression with sustained transgene expression Phase 3: AAV Manufacturing Validation • Apply optimized electroporation conditions to full AAV production • Benchmark performance against PEI-based transfection using reduced plasmid input • Quantitatively assess AAV yield, plasmid efficiency, empty capsid ratio, and batch reproducibility • Demonstrate advancement to TRL 6

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the **primary site for independent biological validation**, including: • Reproduction of transferred electroporation processes in HEK-based producer cells • Quantitative evaluation of cell viability, proliferation, and expression uniformity • Execution of AAV production studies and benchmarking against conventional methods • Analysis of vector genome yield, plasmid utilization efficiency, and empty capsid composition

	• Contribution to scientific publications and technical validation
Expected R&R of Korean SME Partner	The Korean SME partner will act as the **lead industrial developer**, including: • Development and optimization of the flow electroporation platform and cytoprotective process conditions • Integration of hardware, process parameters, and buffer chemistry into a scalable workflow • Core process optimization using DNA, mRNA and plasmid reporter systems • Technology transfer, training, and technical support to the research institute • Data integration, IP creation, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes • Establishment of a validated, manufacturing-ready flow electroporation platform for AAV production • Preservation of post-transfection cell fitness and uniform multi-plasmid delivery • Significant reduction in plasmid consumption and empty capsid formation Industrial and Commercial Impact • Reduced raw material and downstream purification costs • Improved process reproducibility and scalability for AAV manufacturers and CDMOs • Creation of exportable bioprocess equipment, consumables, and manufacturing solutions Academic and Intellectual Property Impact • Generation of peer-reviewed scientific publications • Creation of strategic patent families covering electroporation processes and AAV manufacturing workflows Societal Impact • Support for broader adoption of gene therapies through improved manufacturing efficiency and vector quality • Contribution to global accessibility of treatments for rare and intractable diseases

6. References

1) McColl-Carboni, A. et al. *Analytical characterization of full, intermediate, and empty adeno-associated virus (AAV) capsids.* **Nature Communications**, 2024.
2) Hajba, L. et al. *Recent advances in the analysis of full and empty capsid ratios in adeno-associated virus vectors.* **Molecular Therapy – Methods & Clinical Development**, 2020.
3) Wörner, T. P. et al. *Assessment of genome packaging in adeno-associated viruses using charge-detection mass spectrometry.* **Molecular Therapy – Methods & Clinical Development**, 2022.
4) Li, T. et al. *Determination of full, partial, and empty capsid ratios for adeno-associated virus analysis.* **Analytical Chemistry / Application Note**, 2021.
5) Srivastava, A. et al. *Manufacturing challenges and rational strategies for adeno-associated virus production.* **Journal of Molecular Biology / Bioprocessing Review**, 2021.
6) Park, S. et al. *Enhancing adeno-associated virus production via plasmid ratio optimization.* **Biotechnology Journal**, 2024.

7) Wang, J. et al. *Adeno-associated virus as a delivery vector for gene therapy.* **Signal Transduction and Targeted Therapy**, 2024.
8) Kim, D.-H. *Department of Biomedical Engineering, Johns Hopkins University – Research on engineered cellular systems and cell process engineering.* **Johns Hopkins University Faculty Publications**, selected works.
9) FEMTOBIOMED Inc. *Partitioned Flow Electroporation (PFEP) system for non-viral gene delivery. *Patent application filed in 2022 (pending, publicly disclosed).*